Assessing Comparability with Tertium Comparationis: A linguistical and semantical Approach

The concept of Tertium Comparationis (TC) plays a pivotal role in corpus linguistics, acting as the "common ground" for constructing comparable corpora. Despite its importance, TC remains a subject of debate and lacks a universally accepted definition. This paper aims to address this gap by presenting a comprehensive framework for understanding and applying TC. As preexisting criteria are based on extra-linguistic elements, we address these issues by integrating both measurable linguistic elements and semantic elements through the identification of lexicon-grammatical patterns and the establishment of semantic frames into the formation of the TC.

Firstly, we propose a standardized formalization of the concept, outlining its key criteria and differentiating it from past conceptualizations and methodological approaches. Secondly, we evaluate the strengths and limitations of existing frameworks. This evaluation highlights the need for a more rigorous and nuanced understanding of TC to ensure the validity and reliability of cross-linguistic comparisons. Finally, we introduce a novel dimension and application of TC, extending its role beyond simply establishing a common ground to include a measure of comparability. This paves the way for quantifying similarities between corpora, enabling a more nuanced approach to analyzing language variation.

1. Introduction

This paper explores the challenge of conducting contrastive analyses across different languages using comparable corpora. Establishing direct or semi-direct equivalences requires a robust Tertium Comparationis. Unlike parallel corpora with inherent alignment, comparable corpora necessitate constructing a foundation through a contrastive study of language similarities, variations, and differences. Corpus linguistics offers linguistic elements for this analysis, but also demands adherence to established criteria for comparability (Harris, 1954).

- To be considered comparable, the texts within corpora should share a certain level of similarity, including a common set of named entities (Tóth et al., 2008)

- “To be considered comparable, all texts within corpora should address the same topic” (Preiss, 2012).

Those two parameters emphasize the need for considering the semantic dimension of comparable corpora. Named entities (NER) and document topics are intricately tied to the world of semantics, acting as crucial steppingstones towards deeper comprehension of language. Therefore, those analysis stand as essential tools in unlocking the semantic richness of language. They work in tandem, with NER providing grounding for specific entities and topic modeling revealing broader thematic structures and relationships. Thus, we assume that the concept of Tertium Comparationis must integrate this semantic dimension. Consequently, that notion emerges as a critical element in enabling valid contrastive analyses of comparable corpora at both a linguistic level and semantic one.

2. State-of-the-art

Existing literature on TC is scarce and lacks a comprehensive definition, despite acknowledging its significance in corpus linguistics and the need for clear criteria for its application. Traditionally, TC has been viewed as a "common ground" for comparison (Teubert, 1996; Chesterman, 1998; etc.), enabling the analysis of entities to identify similarities, differences, or equivalence (Andreichuk, 2015). While some propose its use for extracting linguistic equivalences (Connor & Moreno, 2005), the specific criteria employed remain subjective and

dependent on the research focus. The definition of TC varies slightly within these fields. Some scholars (Teubert, 1996; Fix, 2008; etc.) see it primarily as a tool for defining the boundaries of comparable corpora, ensuring homogeneity and authenticity (Loock, 2016). In this view, TC functions solely as a preparatory step during corpus construction.

This paper challenges the traditional understanding of TC as solely a framework for comparable corpora. We propose a paradigm shift, envisioning TC as a dynamic tool capable of facilitating comparisons and quantitatively measuring their validity and strength. This innovant approach has the potential to revolutionize comparable corpora research, ensuring rigorous and replicable findings.

2.1 Preexisting criteria

This section seeks to formalize and validate four recurrent criteria identified as crucial for establishing the Tertium Comparationis (TC). We have designated these criteria as "constant" due to their consistent presence in the literature. Our objective is to demonstrate their significance in shaping and defining a robust TC.

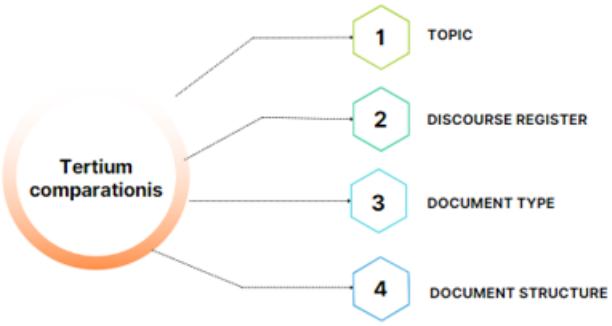

Figure 1. Hierarchy of the constant criteria of Tertium Comparationis

1. The topic centers on identifying and interpreting the central topics and ideas within documents. This analysis focuses on linguistic elements that contribute to the overall meaning and semantic structure of each document. As a foundational step in corpus construction, theme influences the selection of subsequent criteria. Establishing a common theme across comparable corpora facilitates meaningful comparisons and emphasizes similarities or differences more effectively.

2. The discourse register refers to the type of discourse employed in documents. It is defined as a blend of linguistic, stylistic, and pragmatic elements used in a specific communicative situation (Bach, 2020). Beyond its linguistic features, it encompasses socio-cultural aspects that are unique to an individual (Richard, 2009). These elements provide communicative and linguistic insights directly tied to an individual's socio-professional and/or socio-cultural context.

3. The document type characterizes the type of selected documents that will compose the corpus. Similarly, to the topic, having documents with a similar or identical nature ensure the comparison's relevance and reliability. Indeed, it facilitates the comparison of linguistic phenomena (terminology, syntax, communicative situations, etc.) and assures the comparability across languages.

4. The document structure refers to the way paragraphs, titles, and any diagrams are arranged. Choosing documents with similar structures allows for a systematic comparison of corresponding sections across corpora. This consistency enhances the reliability of data and results, making comparisons more reproducible. Additionally, similar structures make it easier to find and extract specific information (Wisniewski et al., 2007; Chung, 2016).

The framework presents a hierarchical structure with cascading conditional relationships among established criteria. Each criterion informs the subsequent one, creating an interdependency. For instance, document types in criterion n°3 depend on the theme in criterion n°1, further refined by the register in criterion n°2. This ensures logical progression and coherence within the criteria set. Yet, it is crucial to note that the current criteria are not precisely and objectively quantifiable.

2.2 Measurable criteria

This section delves into the development of new, quantifiable criteria that are building a robust TC as a tool for measuring comparability across multiple corpora. Focusing on the micro-linguistic level, these criteria require preliminary corpus analysis, serving as an additional layer to the previously established constant criteria. However, to truly capture the nuanced differences and similarities between texts, semantic considerations are also essential. Thus, five measurable criteria are proposed to objectively assess the level of comparability between corpora.

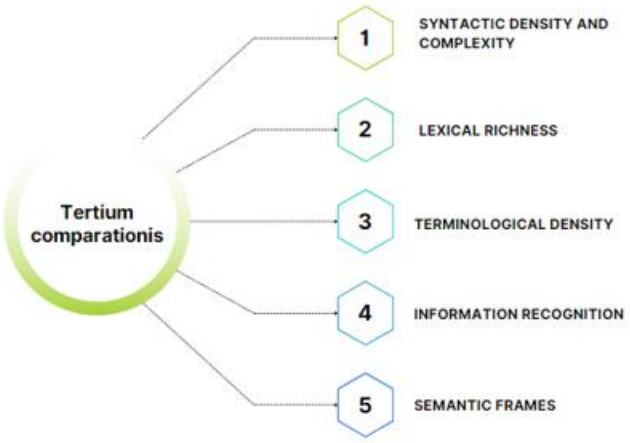

Figure 2. New measurable criteria

1. Syntactic analysis is fundamental for micro-linguistic exploration, examining domain-specific features (Gautier, 2019). Divided into two phases, it involves macroscopic and microscopic analyses of syntactic complexity. The first phase studies sentence size through word count (Bulté & Housen, 2012; Norris & Ortega, 2009), while the second phase delves into language structure, analyzing propositions, constructions, and dependency trees.

2. Lexical richness is assessed through the Type/Token Ratio (TTR). This ratio directly reflects the nature of the document, considering aspects like text type, size, and

specialization. These factors influence the terminology used and its frequency within the document. Essentially, the TTR measures the lexical diversity of a corpus or document, acting as an indicator of both lexical level and typological properties.

3. Terminological density, studied by Ferraresi (2019), distinguishes between specialized terms and common words. It serves two purposes: i) offering a lexical description of the field and its technical terms, and ii) comparing the technicality of specialized terminology as low or high constraint. Analyzing it helps gauge the knowledge expertise required and lays the groundwork for integrating a semantic dimension.

4. Information recognition, spanning diverse languages within a document, leverages semantic analysis at two levels. The macroscopic level compares information content across corpora using semantic techniques like word meaning and thematic structures. In contrast, the microscopic level delves deeper into individual documents, utilizing techniques like discourse analysis, NER and topic-modelling to understand nuanced meaning and thematic variation.

5. Semantic frames, explore how we link language structures (words, phrases) to mental concepts (frames) that guide our interpretation. These frames shape our understanding of language (Fillmore & Baker, 2009). They are connected with the topic criterion. Both share a reliance on conceptual and encyclopedic knowledge specific to a certain domain. Their analysis requires to consider the lexico-grammatical patterns and transferring semantic frames into a multidimensional ontology (Bardzins and al., 2008).

Examining each criterion generates valuable insights into the comparability of the chosen corpora. In the following section we propose methodologies to analyze each of them.

3. Proposed methodologies

We proposed tailored methodologies for each criterion since we recognized their potential interdependence, we assumed that we could extract relevant linguistic information from the corpora for each criterion.

The first step involves calculating the average number of words per sentence to quantify syntactic density. This is formalized as:

$$m_a = \frac{\sum_{i=1}^{n} W_{a_i}}{S_a} = \frac{W_a}{S_a} \tag{1}$$

Reproduce into the second language as:

$$m_b = \frac{\sum_{j=1}^{m} W_{b_j}}{S_b} = \frac{W_b}{S_b} \tag{2}$$

Then,

$$v = \frac{m_a}{m_b} \quad (3)$$

The calculation of the volume of a language is interpreted as follows:

- A represents the reference language.
- B represents the compared language.
- m_a represents the average number of words per sentence in A.
- m_b represents the average number of words per sentence in B.
- S represents the average number of sentences of a document or a corpus.
- W represents the average number of tokens of a document or a corpus.
- v represents the volume of a language.

The second step dives into sentence structure. Dependency trees, generated automatically by SpaCy, and mapping grammatical relationships within documents across languages for easy comparison.

Lexical richness is assessed using the standardized Type-Token Ratio (*s*TTR) applied to the entire corpus. The formula is:

$$sTTR = \frac{TTR}{\sqrt{N}} \quad (4)$$

- TTR refers a ratio relying on the number of types divided by the number of tokens.
- √N refers to the square root of the total number of words in a corpus.

Terminological density can be calculated using tool like SketchEngine and TermoStats. These tools help extract information on term frequency, collocations, and the specialized term of corpora. The latter are then used to create a multilingual corpus and a multidimensional ontology.

Information recognition is based on the use of sentence length that has been previously calculated (cf. syntactic density), a proper segmentation and the use of a dictionary. The Kullback–Leibler (KL) Divergence or relative entropy is expressed as a statistical distance. It serves as a metric to quantify the dissimilarity between a given probability distribution P and a reference probability distribution Q. This distance is used for comparing the divergences between information within several corpora. The formula measures how much the word distribution in the source language (θ_{ws}) differs from the distribution in the target language (θ_{wt}). Additionally, a high result indicates a very different distribution between the source language and the target language, meaning a strong divergence rate between a word x in the source language and a word x in the target language. Conversely, a low result demonstrates a high proximity between the two languages.

$$D_{KL}\,(\theta_{ws} \| \theta_{wt}) = \sum_{i=1}^{|A|} p(d_{si}|w_s) \log \frac{p\,(d_{si}|w_s)}{p\,(d_{ti}|w_t)} \quad (6)$$

Our three-step method analyzes semantic frames within comparable corpora. First, we leverage Topic-Modeling and Named Entity Recognition to identify key semantic categories. This step delves into the conceptual level, identifying broad thematic areas and specific entities. Next, we focus on linguistic patterns by identifying recurrent lexico-grammatical structures within

the corpora. This step delves deeper into the form and function of language used to express these semantic categories. Finally, we transfer the patterns into multilingual ontologies. This final step permits formal comparison and analysis of semantic similarities and differences across corpora.

The methods are applied across all comparable corpora, enabling contrastive analysis of diverse linguistic and semantic elements.

4. Proposed approach

These new criteria will be integrated into a set of metrics that aim to measure the comparability rate for each criterion. Each criterion provides a percentage of similarity between the corpora. This metric will be tested on comparable corpora on the themes of sport in English, French, and Spanish. It aims first to verify the comparability rate of the corpus, i.e., the percentage of similarity between them. Secondly, after defining this percentage, the aimed to align the corpora on a lexical and semantical dimension. We assume that the percentage of comparability serves as an indicator of the percentage of plausible alignment of the corpora.

5. Conclusion

This paper critiques traditional Tertium Comparationis (TC) methods, arguing they lack precise, quantifiable measures for reliable comparisons. Instead, we propose two solutions:

1) Measurable criteria: A set of specific, quantifiable linguistic features to compare across corpora.
2) Analysis methodology: A process to analyze each criterion within comparable corpora.

Our goal is to develop a metric that quantifies the overall comparability between corpora by giving a percentage of comparability. This metric would be more objective and nuanced than current mostly subjective judgements, revealing how closely corpora resemble each other in terms of language use and knowledge representation. Nevertheless, we acknowledge some limitations. Firstly, defining these across diverse languages and domains can be challenging. Secondly, the metric robustness, indeed an extensive testing and validation across different datasets are needed. Finally, incorporating user feedback and domain expertise is crucial for refining such metric. Despite these limitations, this paper proposes a significant step forward: quantifying non-parallel corpus comparability with a comprehensive, data-driven approach to TC. By combining constant and measurable criteria, we aim to empower researchers, linguists, and translators with a powerful tool for objective and reliable cross-linguistic analyses.

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